

# Curriculum vitæ

May 26, 2026

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                            |
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| <b>personalia</b>              | <i>Name:</i> <u>Klas</u> Erik Finn Modin                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                            |
| <b>affiliation</b>             | Department of Mathematical Sciences<br>Chalmers University of Technology and University of Gothenburg<br>Chalmers Tvärgata 3<br>SE-412 96 Göteborg<br>Office phone: +46(0)31-772 35 22<br><i>E-mail:</i> <a href="mailto:klas.modin@chalmers.se">klas.modin@chalmers.se</a><br><i>Web:</i> <a href="https://klasmodin.github.io">klasmodin.github.io</a><br><i>ORCID:</i> <a href="https://orcid.org/0000-0001-6900-1122">0000-0001-6900-1122</a>                      |                                                                                                                                                                                                                                                                            |
| <b>education</b>               | <i>PhD</i> , Mathematics<br>Lund University, Sweden<br>Title: Adaptive Geometric Numerical Integration of Mechanical Systems<br>Supervisors: Claus Führer and Gustaf Söderlind                                                                                                                                                                                                                                                                                         | May 2010                                                                                                                                                                                                                                                                   |
|                                | <i>Master of Science</i> , Mathematics<br>Lund University, Sweden                                                                                                                                                                                                                                                                                                                                                                                                      | Feb 2004                                                                                                                                                                                                                                                                   |
| <b>academic positions</b>      | Professor, <i>Chalmers University of Technology</i><br>Associate Professor, <i>Chalmers University of Technology</i><br>Assistant Professor, <i>Chalmers University of Technology</i><br>Post-doc, <i>University of Toronto</i> , Canada<br>Funded by the <a href="#">Swedish Research Council</a> .<br>Post-doc, <i>Massey University</i> , New Zealand<br>Funded by the <a href="#">Marsden Fund</a> and the <a href="#">Royal Physiographical Society in Lund</a> . | Since Nov 2020<br>Nov 2017–Oct 2020<br>Nov 2013–Oct 2017<br>Jul 2012–Jun 2014<br>Jul 2009–Jan 2012                                                                                                                                                                         |
| <b>other positions</b>         | Numerical Analyst, <i>SKF Sverige AB</i> , Göteborg<br>Employment on a project basis.                                                                                                                                                                                                                                                                                                                                                                                  | 2004–2005                                                                                                                                                                                                                                                                  |
| <b>tutoring<br/>experience</b> | <i>Supervision of Post-docs</i><br><i>Supervision of PhD students</i><br><i>Co-supervision of PhD students</i><br><i>Supervision of master students</i>                                                                                                                                                                                                                                                                                                                | Geir Bogfjellmo (2015–2017)<br>Sagy Ephrati (2023–2025)<br>Eugen Bronasco (2025–today)<br>Kevin Manogue (2026–today)<br>Michael Roop (2021–2026)<br>Erik Jansson (2020–2025)<br>Milo Viviani (2015–2020)<br>currently: 2<br>past: 4<br>Chalmers and GU: 16<br>ENS Paris: 3 |

**teaching***Pedagogical training*

2024. Chalmers EER course “Diversity and Inclusion for Learning” (3 ECTS).  
 2014–2020. Chalmers “Diploma of Higher Education” (17 ECTS).

*Undergraduate teaching*

2019–today. Developing and teaching course on “[Architectural Geometry](#)”.  
 2018–2023. Developing and teaching course on “[Scientific Visualization](#)”.  
 2014–today. Basic calculus courses at Chalmers. (Teacher and examiner.)  
 2010–2012. Various mathematics courses at Massey. (Teacher and examiner.)

*Post-graduate teaching*

2025. Mini-course on “[Information geometry of diffeomorphism groups](#)” at the Erwin Schrödinger Institute, Vienna, Austria, February 3–7, 2025.  
 2024. Course on “[Numerical Methods for ODEs](#)” at Chalmers and GU.  
 2023. Mini-course on “[Geometric Hydrodynamics on Flatland](#)” at the University of Santiago de Compostella, Spain, September 4–8, 2023.  
 2018. Mini-course on “[Geometric Hydrodynamics](#)” at the University of Coimbra, Portugal, December 6–8, 2018.  
 2013. Course on “Geometric Numerical Integration” at Chalmers and GU.

*Written lecture notes*

2013. “[Geometric Mechanics and Geometric Integration](#)”.

**workshop****organizer at****research institutes**

|                                                      |          |
|------------------------------------------------------|----------|
| Institute Mittag–Leffler (IML), Stockholm, Sweden    | Jul 2027 |
| Sophus Lie Conference Center for Mathematics, Norway | Jul 2024 |
| Banff International Research Station (BIRS), Canada  | Nov 2023 |
| Institute Mittag–Leffler (IML), Stockholm, Sweden    | Jul 2018 |

**invitations to****research institutes**

|                                                                         |              |
|-------------------------------------------------------------------------|--------------|
| Mathematisches Forschungsinstitut Oberwolfach (MFO), Germany            | Apr 2027     |
| Mathematisches Forschungsinstitut Oberwolfach (MFO), Germany            | Jan 2027     |
| Mathematisches Forschungsinstitut Oberwolfach (MFO), Germany            | May 2026     |
| International Centre for Mathematical Sciences (ICMS), Edinburgh, UK    | Jun 2025     |
| Gran Sasso Science Institute (GSSI), L’Aquila, Italy                    | May 2025     |
| Erwin Schrödinger Institute (ESI), Vienna, Austria                      | Feb 2025     |
| Centre International de rencontres mathématiques (CIRM), Luminy, France | May 2024     |
| Institute Mittag–Leffler (IML), Stockholm, Sweden                       | Nov 2023     |
| Max Planck Institute (MPL), Leipzig, Germany                            | Apr 2023     |
| Simon Center for Geometry and Physics (SCGP), New York, USA             | Jul 2022     |
| International Center for Mathematical Science (ICMS), Edinburgh, UK     | Apr 2021     |
| Mathematisches Forschungsinstitut Oberwolfach (MFO), Germany            | Mar 2021     |
| Fields Institute (FI), Toronto, Canada                                  | Sep 2020     |
| Princeton Center of Theoretical Sciences (PCTS), Princeton, USA         | Feb 2020     |
| Hausdorff Research Institute (HIM), Bonn, Germany                       | Nov 2019     |
| Isaac Newton Insitute (INI), UK                                         | Nov 2019     |
| Banff International Research Station (BIRS), Canada                     | Dec 2018     |
| Isaac Newton Insitute (INI), UK                                         | Nov 2017     |
| Mathematisches Forschungsinstitut Oberwolfach (MFO), Germany            | Mar 2016     |
| Erwin Schrödinger Institute (ESI), Vienna, Austria                      | Jan 2015     |
| Simon Center for Geometry and Physics (SCGP), New York, USA             | May 2014     |
| Fields Institute (FI), Toronto, Canada                                  | Jul–Aug 2012 |

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| <b>outreach activities</b> | Public lectures at Vetenskapsfestivalen (International Science Festival Gothenburg) and at Bokmässan (Gothenburg Book Fair)                                                                               | 2014–today |
|                            | “Library Live” at Chalmers Library, discussion between Julia Ravanis and Klas Modin on the life of Emmy Noether                                                                                           | May 2025   |
|                            | Interview in the cultural magazine “Glänta” (in Swedish)                                                                                                                                                  | Aug 2025   |
|                            | Short-movie about our research, made by KAW for outreach purposes                                                                                                                                         | Jan 2026   |
|                            | “Glänta i foajén: Rymdens oändlighet” at Folkteatern Göteborg, panel discussion with Susanne Aalto, Balsam Karam, Klas Modin, and Julia Ravanis                                                           | Apr 2026   |
| <b>other achievements</b>  | Co-founder of the MedTech company <a href="#">Navari Surgical</a> for intraoperative augmented reality guidance during laparoscopic surgery. It was placed on <a href="#">IVA’s top 100 list</a> in 2023. |            |
|                            | Initiator and main developer for the open-source project <a href="#">QUFLOW</a> for structure preserving discretizations in hydrodynamics.                                                                |            |

## Selection of honours and grants

- 2026. *Wallenberg PoC Grant*, Knut and Alice Wallenberg Foundation (KAW).
- 2026. *Plenary speaker*, Foundations of Computational Mathematics (FoCM), Vienna, July 2026.
- 2025. *Göran Gustafsson Award in Mathematics*, Göran Gustafsson Foundation.
- 2024. *Wallenberg Academy Fellow (extension)*, Knut and Alice Wallenberg Foundation (KAW).
- 2024. *International post-doc recruitment grant*, Knut and Alice Wallenberg Foundation (KAW).
- 2022. *Project Grant*, Swedish Research Council (VR).
- 2019. *Wallenberg Academy Fellow*, Knut and Alice Wallenberg Foundation (KAW).
- 2017. *Starting Grant*, Swedish Research Council (VR).
- 2015. *International post-doc recruitment grant*, Knut and Alice Wallenberg Foundation (KAW).
- 2015. *Stenbäckska Stipendiet*, Finnish Society of Sciences and Letters.
- 2015. *Marie Skłodowska-Curie Individual Fellowship*, EU Horizon 2020.
- 2015. *Transition Grant*, Swedish Foundation for International Cooperation in Research and Higher Education (STINT).
- 2013. *Ingvar Carlsson Award*, Swedish Foundation of Strategic Research (SSF).
- 2012. *International Post-doc grant*, Swedish Research Council (VR).
- 2010. *Post-doctoral scholarship* Royal Physiographic Society in Lund.
- 2009. *Travel scholarship* Royal Swedish Academy of Science (KVA).
- 2007. *Young researcher scholarship* Royal Physiographic Society in Lund.

## Peer-reviewed publications

For updates and other publications, see [klasmodin.github.io/publications](https://klasmodin.github.io/publications)

- 60. Cifani, P., Modin, K., Pagliantini, C. & Viviani, M. Isospectral symplectic Runge–Kutta schemes as Lie group methods. *J. Comput. Dyn.* **14**, 67–73 (2026).
- 59. Lichtenfelz, L., Modin, K. & Preston, S. C. Ricci curvature for hydrodynamics on the sphere. *Comm. Math. Phys.* **407**, 37 (2026).
- 58. Modin, K. & Roop, M. Structure-preserving long-time simulations of turbulence in magnetized ideal fluids. *J. Plasma Phys.* **92**, E30 (2 2026).

57. Modin, K. & Viviani, M. A brief introduction to matrix hydrodynamics. *J. Comput. Dyn.* **14**, 17–35 (2026).
56. Modin, K. & Viviani, M. Two-dimensional fluids via matrix hydrodynamics. *Arch. Rational Mech. Anal.* **250**, 10 (2026).
55. Ephrati, S., Jansson, E. & Modin, K. On spectral scaling laws for averaged turbulence on the sphere. *Phys. D* **481**, 134808 (2025).
54. Jansson, E., Krook, J., Modin, K. & Öktem, O. Geometric shape matching for recovering protein conformations from single-particle Cryo-EM data. *SIAM J. Imaging Sci.* **18**, 2509–2546 (2025).
53. Jansson, E. & Modin, K. Sub-Riemannian landmark matching and its interpretation as residual neural networks. *J. Comput. Dyn.* **12**, 467–490 (3 2025).
52. Modin, K. & Preston, S. C. Zeitlin’s model for axisymmetric 3-D Euler equations. *Nonlinearity* **38**, 025008 (2025).
51. Modin, K. & Roop, M. Spatio-temporal Lie-Poisson discretization for incompressible magnetohydrodynamics on the sphere. *IMA J. Numer. Anal.*, draf024 (2025).
50. Modin, K. & Suri, A. Geodesic interpretation of the global quasi-geostrophic equations. *Calc. Var. Partial Differential Equations* **65**, 17 (2025).
49. Jansson, E. & Modin, K. Convergence of the vertical gradient flow for the Gaussian Monge problem. *J. Comput. Dyn.* **11**, 1–9 (2024).
48. Khesin, B., Modin, K. & Volk, L. Simple unbalanced optimal transport. *Int. Math. Res. Not.* **2024**, 8839–8855 (10 2024).
47. Modin, K. On the geometry and dynamical formulation of the Sinkhorn algorithm for optimal transport. *J. Comput. Dyn.* **11**, 442–452 (2024).
46. Modin, K. & Perrot, M. Eulerian and Lagrangian stability in Zeitlin’s model of hydrodynamics. *Comm. Math. Phys.* **405**, 177 (2024).
45. Cifani, P., Viviani, M. & Modin, K. An efficient geometric method for incompressible hydrodynamics on the sphere. *J. Comput. Phys.* **473**, 111772 (2023).
44. Khesin, B. & Modin, K. The Toda flow as a porous medium equation. *Comm. Math. Phys.* **401**, 1879–1898 (2023).
43. Maurelli, M., Modin, K. & Schmeding, A. Incompressible Euler equations with stochastic forcing: a geometric approach. *Stochastic Process. Appl.* **159**, 101–148 (2023).
42. Balehowsky, T., Karlsson, C.-J. & Modin, K. Shape analysis via gradient flows on diffeomorphism groups. *Nonlinearity* **36**, 862 (2022).
41. Cifani, P., Viviani, M., Luesink, E., Modin, K. & Geurts, B. Casimir preserving spectrum of two-dimensional turbulence. *Phys. Rev. Fluids* **7**, L082601 (2022).
40. Modin, K. & Viviani, M. Canonical scale separation in two-dimensional incompressible hydrodynamics. *J. Fluid Mech.* **943**, A36 (2022).
39. Khesin, B., Misiolek, G. & Modin, K. Geometric hydrodynamics and infinite-dimensional Newton’s equations. *Bull. Amer. Math. Soc.* **58**, 377–442 (2021).
38. Modin, K. & Viviani, M. Integrability of point-vortex dynamics via symplectic reduction: a survey. *Arnold Math. J.* **7**, 357–385 (2021).
37. Bauer, M. & Modin, K. Semi-invariant Riemannian metrics in hydrodynamics. *Calc. Var. Partial Differential Equations* **59**, 65 (2020).
36. Modin, K. & Verdier, O. What makes nonholonomic integrators work? *Numer. Math.* **145**, 405–435 (2020).
35. Modin, K. & Viviani, M. A Casimir preserving scheme for long-time simulation of spherical ideal hydrodynamics. *J. Fluid Mech.* **884**, A22 (2020).

34. Modin, K. & Viviani, M. Lie-Poisson methods for isospectral flows. *Found. Comput. Math.* **20**, 889–921 (2020).
33. Benn, J., Marsland, S., McLachlan, R., Modin, K. & Verdier, O. Currents and finite elements as tools for shape space. *J. Math. Imaging Vis.* **61**, 1197–1220 (2019).
32. Hellsvik, J. *et al.* General method for atomistic spin-lattice dynamics with first-principles accuracy. *Phys. Rev. B* **99**, 104302 (2019).
31. Khesin, B., Misiolek, G. & Modin, K. Geometry of the Madelung transform. *Arch. Rational Mech. Anal.* **234**, 549–573 (2019).
30. Bogfjellmo, G., Modin, K. & Verdier, O. A Numerical Algorithm for C2-splines on Symmetric Spaces. *SIAM J. Numer. Analysis* **56**, 2623–2647 (2018).
29. Khesin, B., Misiolek, G. & Modin, K. Geometric Hydrodynamics via Madelung Transform. *Proc. Natl. Acad. Sci. USA* **115**, 6165–6170 (2018).
28. Modin, K., Nachman, A. & Rondi, L. A Multiscale Theory for Image Registration and Nonlinear Inverse Problems. *Adv. Math.* **346**, 1009–1066 (2018).
27. Bauer, M., Joshi, S. & Modin, K. *Diffeomorphic random sampling using optimal information transport in Nielsen F., Barbaresco F. (eds) Geometric Science of Information. GSI 2017. Lecture Notes in Computer Science, vol 10589. Springer* (2017).
26. Bauer, M., Joshi, S. & Modin, K. On Geodesic Completeness of Riemannian Metrics on Smooth Probability Densities. *Calc. Var. Partial Differential Equations* **56**, 113 (2017).
25. McLachlan, R., Modin, K., Munthe-Kaas, H. & Verdier, O. Butcher series: A story of rooted trees and numerical methods for evolution equations. *Asia Pacific Mathematics Newsletter* **7**, 1–11 (2017).
24. McLachlan, R., Modin, K. & Verdier, O. A minimal-variable symplectic integrator on spheres. *Math. Comp.* **86**, 2325–2344 (2017).
23. Modin, K. Geometry of Matrix Decompositions Seen Through Optimal Transport and Information Geometry. *J. Geom. Mech.* **9**, 335–390 (2017).
22. McLachlan, R., Modin, K., Munthe-Kaas, H. & Verdier, O. B-series methods are exactly the affine equivariant methods. *Numer. Math.* **133**, 599–622 (2016).
21. McLachlan, R., Modin, K. & Verdier, O. Geometry of discrete-time spin systems. *J. Nonlin. Sci.* **26**, 1507–1523 (2016).
20. McLachlan, R., Modin, K. & Verdier, O. Symmetry reduction for central force problems. *Eur. J. Phys.* **37**, 0055003 (2016).
19. Bauer, M., Joshi, S. & Modin, K. Diffeomorphic density matching by optimal information transport. *SIAM J. Imaging Sci.* **8**, 1718–1751 (2015).
18. McLachlan, R., Modin, K. & Verdier, O. Collective Lie-Poisson integrators on  $R^3$ . *IMA. J. Num. Anal.* **35**, 546–560 (2015).
17. Modin, K. Generalized Hunter-Saxton equations, optimal information transport, and factorization of diffeomorphisms. *J. Geom. Anal.* **25**, 1306–1334 (2015).
16. Rottman, C., Bauer, M., Modin, K. & Joshi, S. *Weighted Diffeomorphic Density Matching with Applications to Thoracic Image Registration in Proc. 5th MICCAI Workshop on Mathematical Foundations of Computational Anatomy (MFCA), Munich, Germany, October 9* (2015).
15. Marsland, S., McLachlan, R., Modin, K. & Perlmutter, M. On conformal variational problems and free boundary continua. *J. Phys. A* **47**, 145204 (2014).
14. McLachlan, R., Modin, K. & Verdier, O. Collective symplectic integrators. *Nonlinearity* **27**, 1525–1542 (2014).
13. McLachlan, R., Modin, K. & Verdier, O. Symplectic integrators for spin systems. *Phys. Rev. E* **89**, 061301 (2014).

12. McLachlan, R., Modin, K., Verdier, O. & Wilkins, M. Geometric Generalisations of SHAKE and RATTLE. *Found. Comput. Math.* **14**, 339–370 (2014).
11. Marsland, S., McLachlan, R., Modin, K. & Perlmutter, M. Geodesic Warps by Conformal Mappings. *Int. J. Comput. Vis.* **105**, 144–154 (2013).
10. McLachlan, R., Modin, K., Verdier, O. & Wilkins, M. Symplectic integrators for index 1 constraints. *SIAM J. Sci. Comput.* **35**, A2150–A2162 (2013).
9. Modin, K. & Verdier, O. Integrability of Nonholonomically Coupled Oscillators. *Discrete Contin. Dyn. Syst.* **34**, 1121–1130 (2013).
8. Marsland, S., McLachlan, R., Modin, K. & Perlmutter, M. *On a Geodesic Equation for Planar Conformal Template Matching* in *Proc. MFCA'11* (2011).
7. Modin, K., Perlmutter, M., Marsland, S. & McLachlan, R. On Euler-Arnold Equations and Totally Geodesic Subgroups. *J. Geom. Phys.* **61**, 1446–1461 (2011).
6. Modin, K. & Söderlind, G. Geometric Integration of Hamiltonian Systems Perturbed by Rayleigh Damping. *BIT Num. Math.* **51**, 977–1007 (2011).
5. Modin, K. Time-transformation and reversibility of Nambu-Poisson systems. *J. Gen. Lie Theory Appl.* **3**, 39–52 (2009).
4. Modin, K. On explicit adaptive symplectic integration of separable Hamiltonian systems. *J. Mult. Body Mech.* **222**, 1464–1493 (2008).
3. Modin, K., Fritzson, D. & Führer, C. *Semiexplicit Numerical Integration by Splitting with Application to Dynamic Multibody Problems with Contacts* in *Proceedings of The 48th Scandinavian Conference on Simulation and Modeling (SIMS 2007)*, Linköping University Electronic Press (2007).
2. Modin, K. & Führer, C. Time-step adaptivity in variational integrators with application to contact problems. *ZAMM Z. Angew. Math. Mech.* **86**, 785–794 (2006).
1. Modin, K., Fritzson, D., Führer, C. & Söderlind, G. *A new class of variable step-size methods for multi-body dynamics* in *Proceedings of Multibody Dynamics 2005, ECCOMAS Thematic Conference, Madrid, June 21-24* (2005).